



Oral Dwellers

Here's a closer genetic look at the bacteria living inside the human mouth. <http://digit.in/jan21-21>



Pelting Tomatoes On Jitteriness

Tomatoes are being genetically modified to offer an affordable source of Parkinson's disease drugs. <http://digit.in/jan21-22>

Advancements in Microrobotics

Recent research around making small robots more versatile and powerful

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lot of the promise of nanotechnology can actually be realised on the micro scale, and there is constant

development in microrobotics, with new systems demonstrating novel capabilities. In the last few months, researchers have demonstrated fast morphing robots, tiny robots controlled by lasers, and robots that can tumble their way through the guts of humans.

ORIGAMI FOR SHAPE SHIFTING ROBOTS

Researchers from Ohio State University and Georgia Tech have combined magnetism with the ancient art of origami, to create a responsive system with untethered controls. The magnetic actuation removes both the power source and the controller from the robot, allowing for the entire system to scale. Proposed use cases for the system include shape shifting robots, satellites and biomedical devices. Most of the conventional approaches for using origami techniques to allow robots to change shape depend on a complex network of wires and actuators.

To demonstrate the system, the researchers created cylindrical tubes in an origami pattern known as the



Credit: Georgia Tech

Kresling pattern, made up of identical triangular cells. One of the researchers, Glaucio Paulino explains why this pattern is of interest in robotics, "the Kresling pattern offers a very rich design space, which was crucial in coupling its mechanical response with magnetically responsive materials to achieve on-demand, untethered actuation, including our multifunctional origami for digital computing."

The pattern allows the structure to expand or compress, and fold under load. Think of a single unit in the system as a cylinder with an expanded or compressed state. The walls of the cylinder are made up of

the Kresling pattern. These cylindrical units can be made to expand or compress rapidly, and exist only momentarily in between the two states. By combining many of these cylindrical units together, it is possible to create complex robots. Additionally, various characteristics of each of the cells or units can be controlled, all through magnetism.

The speed and direction of the materials deployment was controlled by a magnetic field. The researchers then attached a magnetised plate to each of the triangular cells in the Kresling pattern. This allowed the researchers to arbitrarily manipulate the